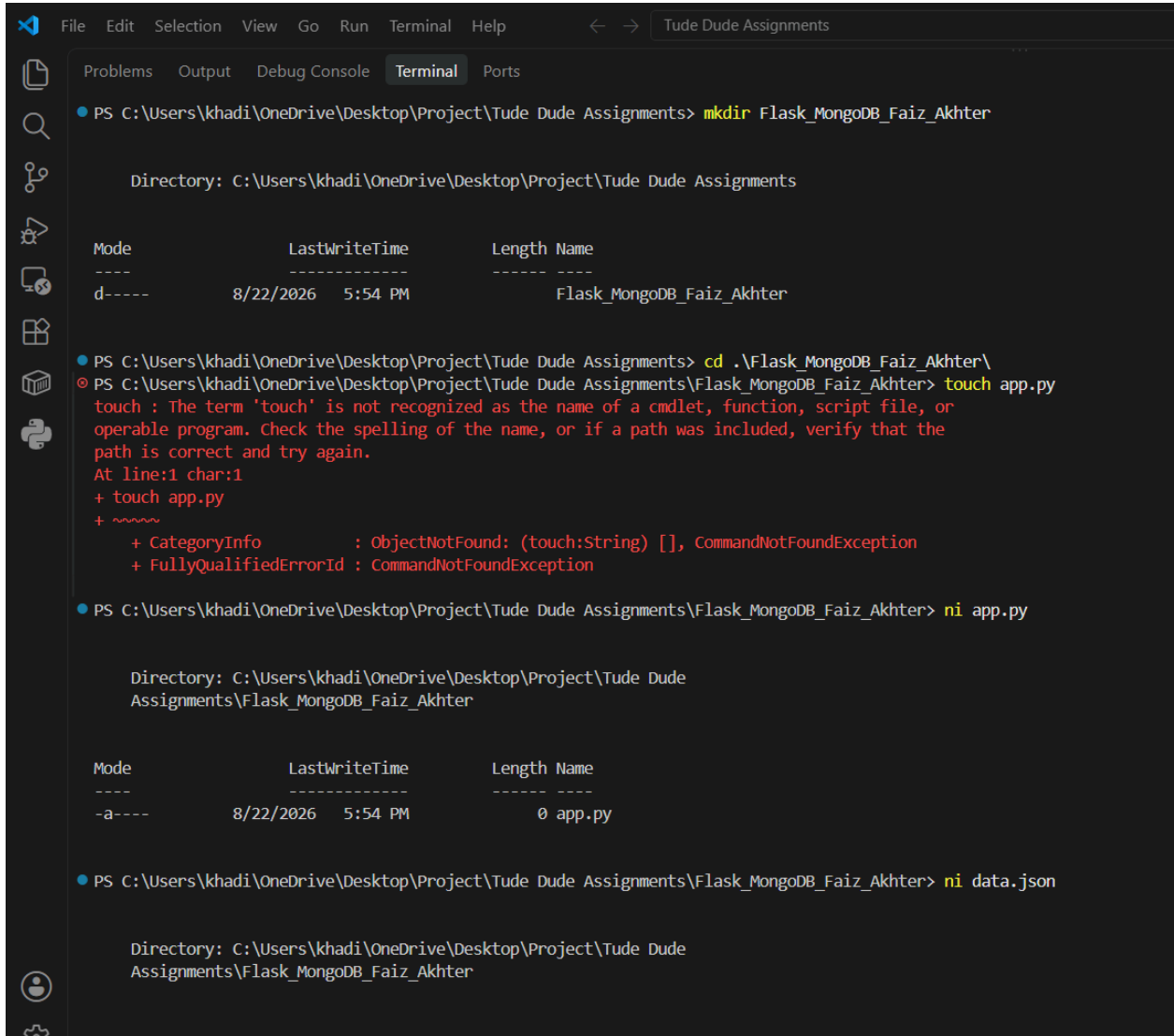


Build a Flask application with a JSON API route and a frontend form that stores submitted data in MongoDB Atlas, with proper success and error handling.

Task 1: JSON API Route

Step 1: creating folders



```
File Edit Selection View Go Run Terminal Help
Tude Dude Assignments

Problems Output Debug Console Terminal Ports

PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments> mkdir Flask_MongoDB_Faiz_Akhter

Directory: C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments

Mode                LastWriteTime         Length Name
----                -
d-----            8/22/2026   5:54 PM                Flask_MongoDB_Faiz_Akhter

PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments> cd .\Flask_MongoDB_Faiz_Akhter\
PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> touch app.py
touch : The term 'touch' is not recognized as the name of a cmdlet, function, script file, or
operable program. Check the spelling of the name, or if a path was included, verify that the
path is correct and try again.
At line:1 char:1
+ touch app.py
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (touch:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> ni app.py

Directory: C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude
Assignments\Flask_MongoDB_Faiz_Akhter

Mode                LastWriteTime         Length Name
----                -
-a-----            8/22/2026   5:54 PM                0 app.py

PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> ni data.json

Directory: C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude
Assignments\Flask_MongoDB_Faiz_Akhter
```

```
Problems Output Debug Console Terminal Ports powershell - Flask_MongoDB_Faiz_Akhter + - [ ] [ ] ... | [ ] [ ] >

PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> ni app.py
• PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> ni data.json

Directory: C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter

Mode                LastWriteTime         Length Name
----                -
-a-----          8/22/2026   5:55 PM              0 data.json

• PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> ni requirement.txt

Directory: C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter

Mode                LastWriteTime         Length Name
----                -
```

```
• PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> ni requirement.txt

Directory: C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter

Mode                LastWriteTime         Length Name
----                -
-a-----          8/22/2026   5:55 PM              0 requirement.txt

○ PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> |
```

```
Mode                LastWriteTime         Length Name
----                -
-a-----          8/22/2026   5:58 PM              0 .env
-a-----          8/22/2026   5:54 PM              0 app.py
-a-----          8/22/2026   5:55 PM              0 data.json
-a-----          8/22/2026   5:55 PM              0 requirement.txt

○ PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_Akhter> |
```

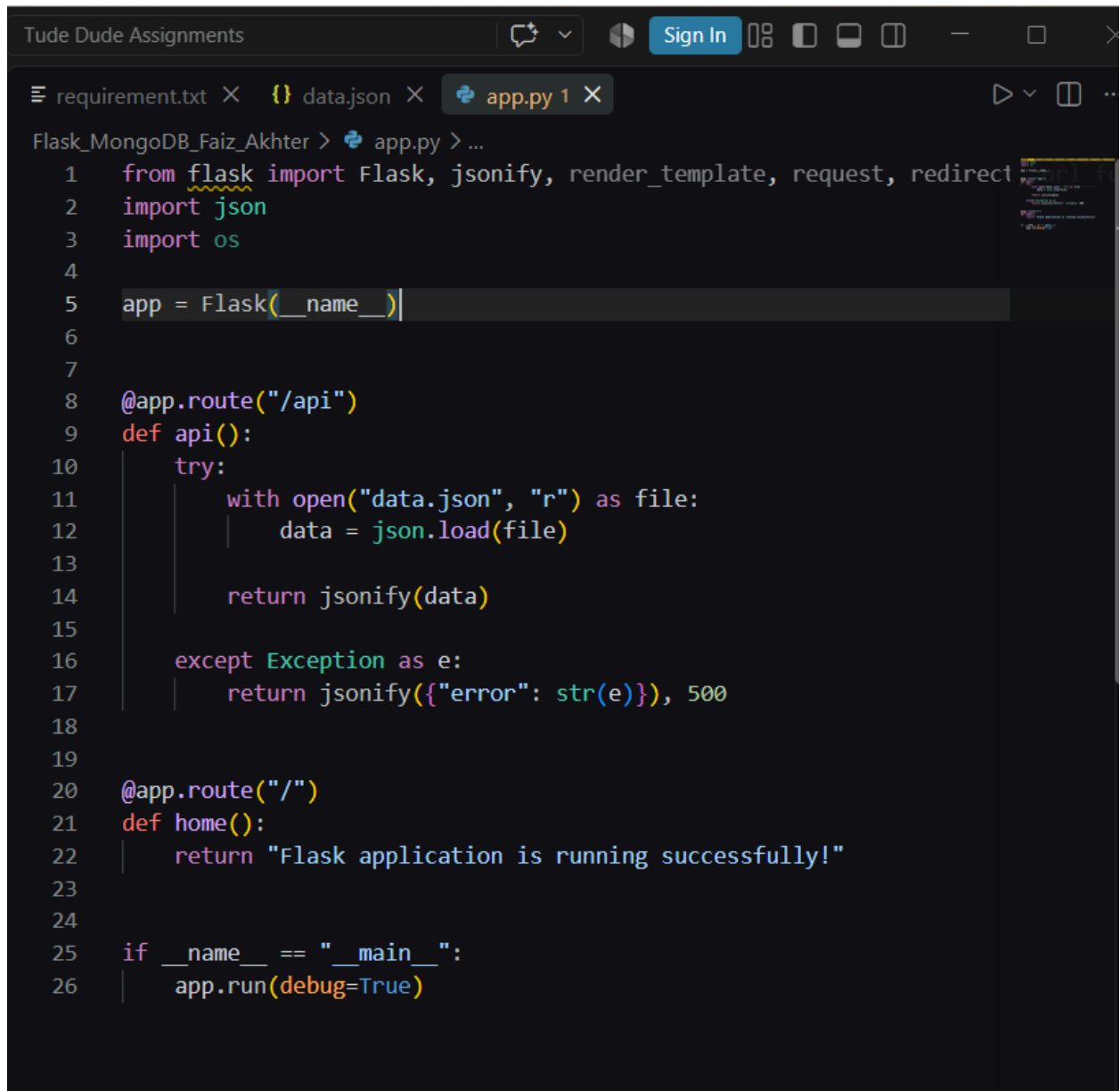
Step 2: setting up python Flask environment

```
PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_
● Akhter> python -m venv venv
PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoDB_Faiz_
● Akhter> .\venv\Scripts\Activate.ps1
(venv) PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoD
● B_Faiz_Akhter> Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
(venv) PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoD
● B_Faiz_Akhter> .\venv\Scripts\Activate.ps1
○ (venv) PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoD
B_Faiz_Akhter> |
```

```
Using cached dnspython-2.8.0-py3-none-any.whl (331 kB)
Using cached python_dotenv-1.2.3-py3-none-any.whl (22 kB)
Using cached blinker-1.9.0-py3-none-any.whl (8.5 kB)
Using cached click-8.4.2-py3-none-any.whl (119 kB)
Using cached itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Using cached markupsafe-3.0.3-cp314-cp314-win_amd64.whl (15 kB)
Using cached werkzeug-3.1.8-py3-none-any.whl (226 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: python-dotenv, markupsafe, itsdangerous, dnspython, c
olorama, blinker, werkzeug, pymongo, jinja2, click, Flask
Successfully installed Flask-3.1.3 blinker-1.9.0 click-8.4.2 colorama-0.4.6 dnspytho
n-2.8.0 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 pymongo-4.17.0 python-doten
v-1.2.3 werkzeug-3.1.8

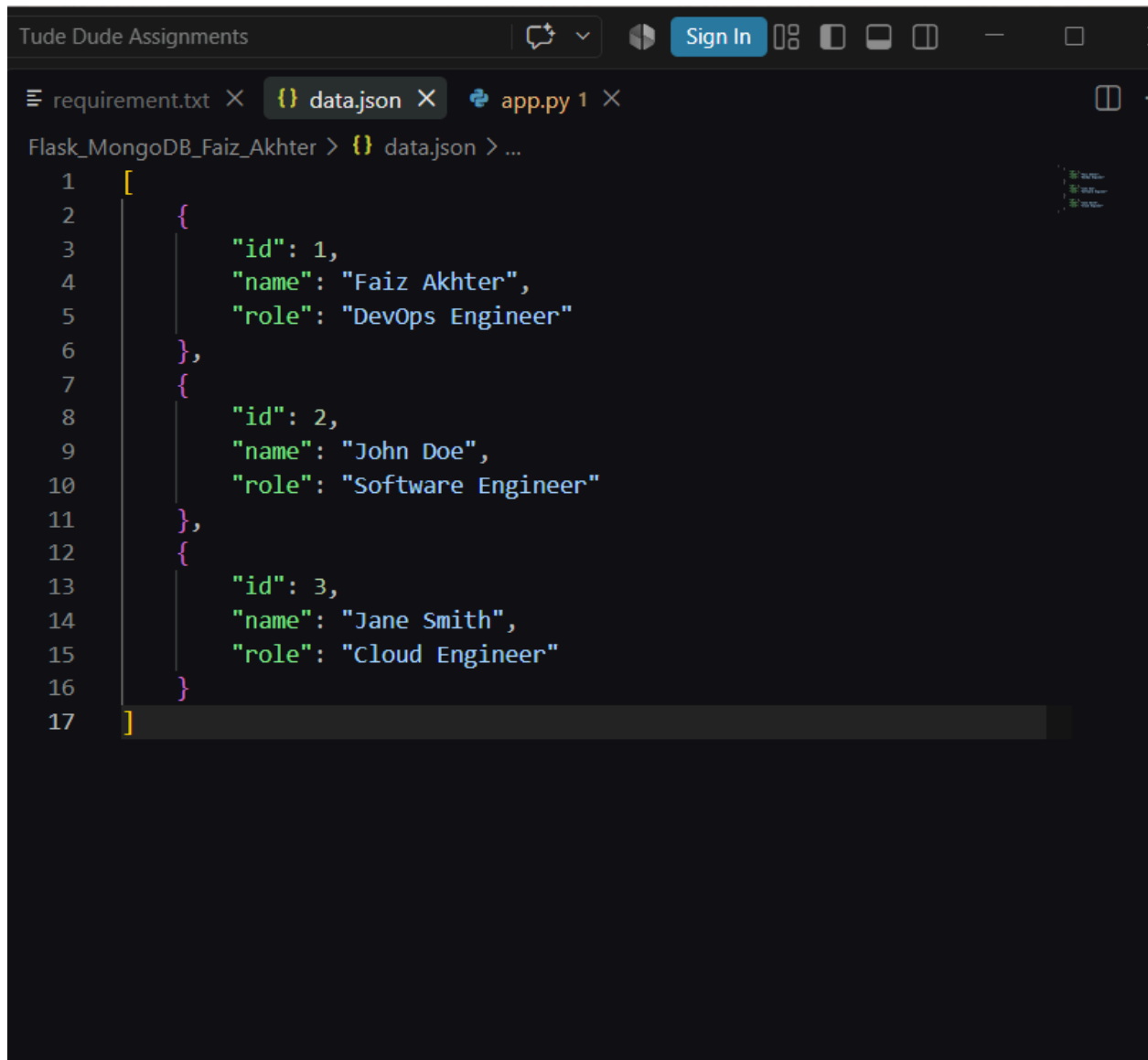
[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoD
B_Faiz_Akhter> pip freeze > requirements.txt
(venv) PS C:\Users\khadi\OneDrive\Desktop\Project\Tude Dude Assignments\Flask_MongoD
B_Faiz_Akhter> |
```

Step 3: writing json and [app.py](#) files



The screenshot shows a code editor window titled 'Tude Dude Assignments'. The editor has three tabs: 'requirement.txt', 'data.json', and 'app.py 1'. The 'app.py' tab is active, showing the following Python code:

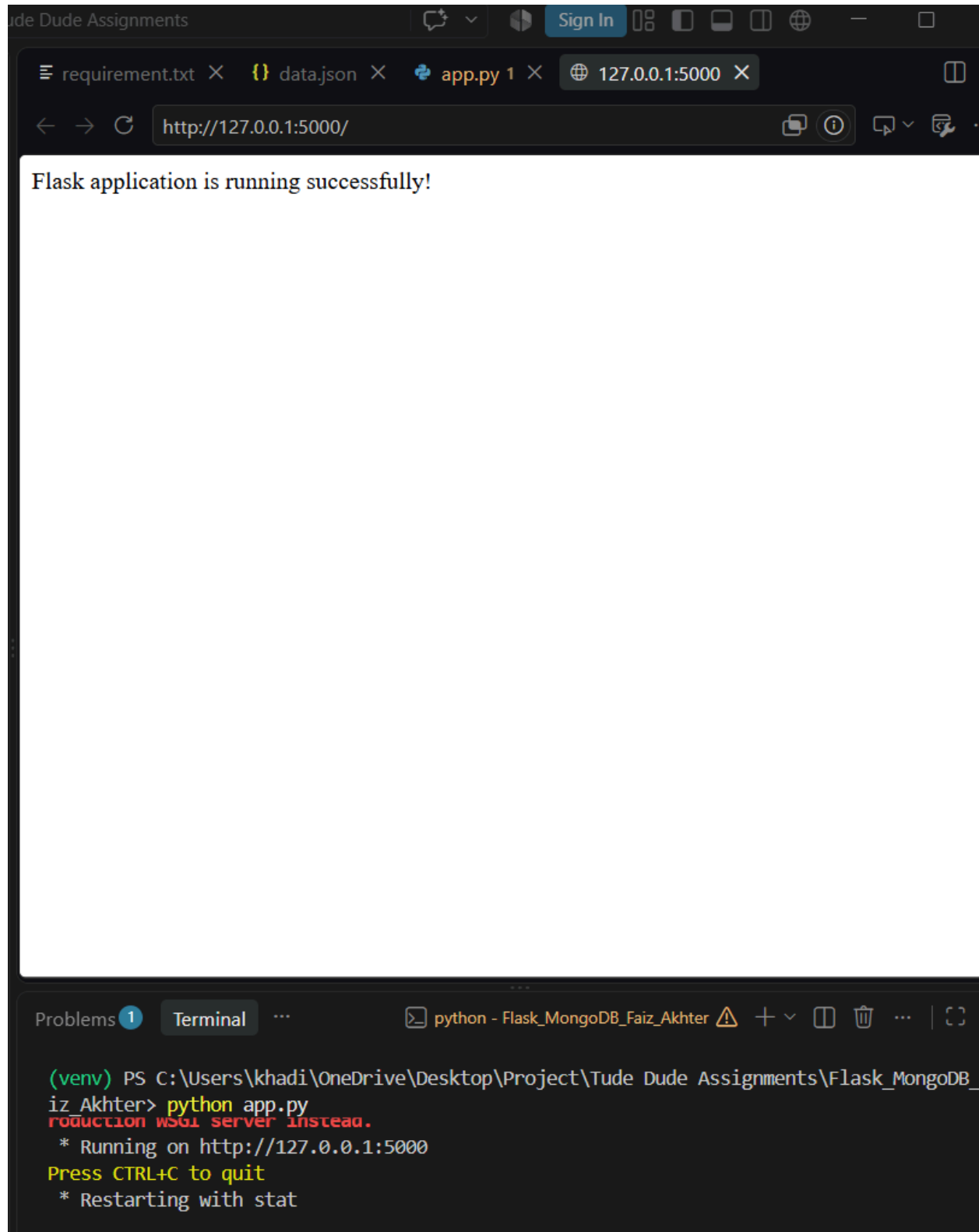
```
Flask_MongoDB_Faiz_Akhter > app.py > ...
1  from flask import Flask, jsonify, render_template, request, redirect
2  import json
3  import os
4
5  app = Flask(__name__)
6
7
8  @app.route("/api")
9  def api():
10     try:
11         with open("data.json", "r") as file:
12             data = json.load(file)
13
14         return jsonify(data)
15
16     except Exception as e:
17         return jsonify({"error": str(e)}), 500
18
19
20 @app.route("/")
21 def home():
22     return "Flask application is running successfully!"
23
24
25 if __name__ == "__main__":
26     app.run(debug=True)
```



The screenshot shows a code editor window with the title "Tude Dude Assignments". The editor has three tabs: "requirement.txt", "data.json", and "app.py 1". The "data.json" tab is active, showing a JSON array of three objects. The first object has "id": 1, "name": "Faiz Akhter", and "role": "DevOps Engineer". The second object has "id": 2, "name": "John Doe", and "role": "Software Engineer". The third object has "id": 3, "name": "Jane Smith", and "role": "Cloud Engineer". The code is syntax-highlighted with colors: strings in green, numbers in blue, and punctuation in pink. The line numbers 1 through 17 are visible on the left side of the editor.

```
1  [  
2    {  
3      "id": 1,  
4      "name": "Faiz Akhter",  
5      "role": "DevOps Engineer"  
6    },  
7    {  
8      "id": 2,  
9      "name": "John Doe",  
10     "role": "Software Engineer"  
11   },  
12   {  
13     "id": 3,  
14     "name": "Jane Smith",  
15     "role": "Cloud Engineer"  
16   }  
17 ]
```

After this run the command `python app.py`



It is successfully running on port 5000

Now! Checking out api page....

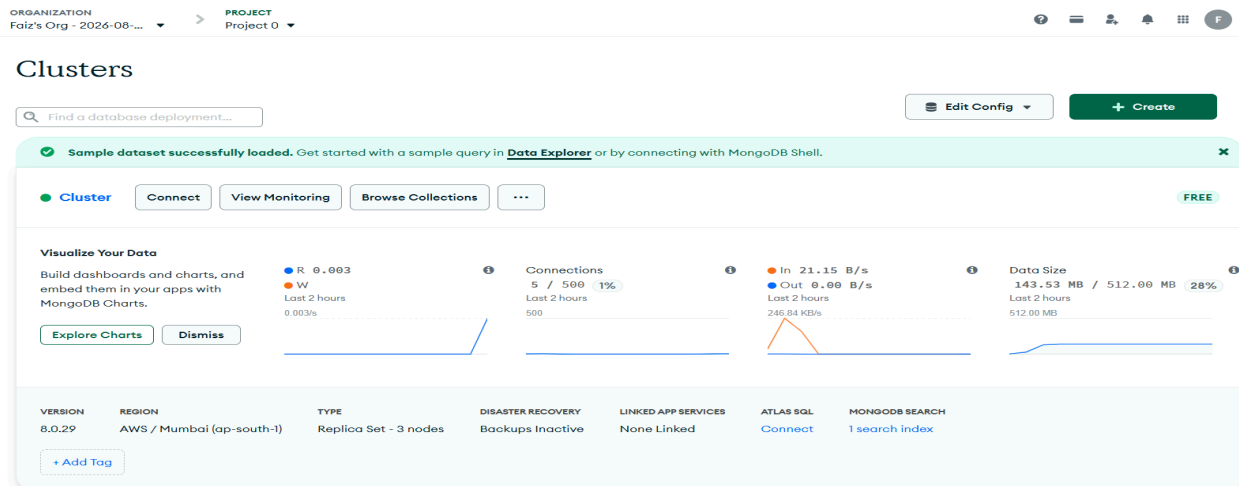
A screenshot of a web browser window. The address bar shows the URL 'http://127.0.0.1:5000/api'. Below the address bar, there is a tab labeled 'Pretty-print'. The main content area of the browser displays a JSON array of three objects, each representing a person with an id, name, and role. The JSON is formatted with syntax highlighting and collapsible nodes.

```
[
  {
    "id": 1,
    "name": "Faiz Akhter",
    "role": "DevOps Engineer"
  },
  {
    "id": 2,
    "name": "John Doe",
    "role": "Software Engineer"
  },
  {
    "id": 3,
    "name": "Jane Smith",
    "role": "Cloud Engineer"
  }
]
```

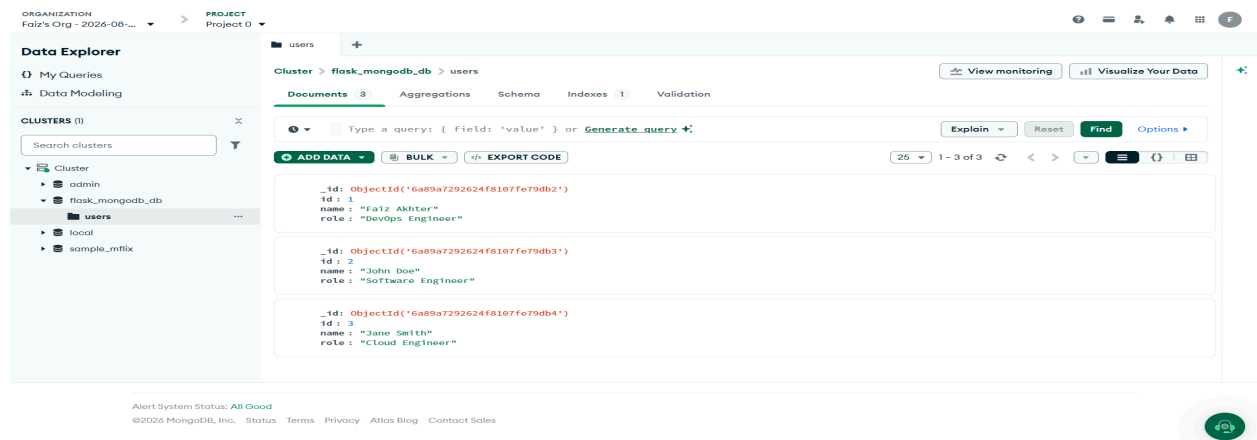
Yes it is visible !

Task 2: Frontend Form with MongoDB Atlas

Step 4: making MongoDB and connect it with my flask api



Yes, it is running and my api is successfully working



It is successfully reflecting the data from json

```
data.json x pymongo.errors.ServerSelection... x .env x 127.0.0.1
Flask_MongoDB_Faiz_Akhter > {} data.json > ...
1  [
2      {
3          "id": 1,
4          "name": "Faiz Akhter",
5          "role": "DevOps Engineer"
6      },
7      {
8          "id": 2,
9          "name": "John Doe",
10         "role": "Software Engineer"
11     },
12     {
13         "id": 3,
14         "name": "Jane Smith",
15         "role": "Cloud Engineer"
16     }
17 ]
```

Step 5: creating frontend for data

127.0.0.1:5000

FLASK x MONGODB ATLAS

Build Your Cloud Data Profile

Submit your professional details securely through our Flask application and store your information directly in MongoDB Atlas.

- Cloud Connected**
Your data is securely stored in MongoDB Atlas.
- Fast Submission**
Submit your information through the Flask API.
- API Powered**
Access imported database records through the API.

Submit Your Data

Complete the form below to save your details.

Full Name *

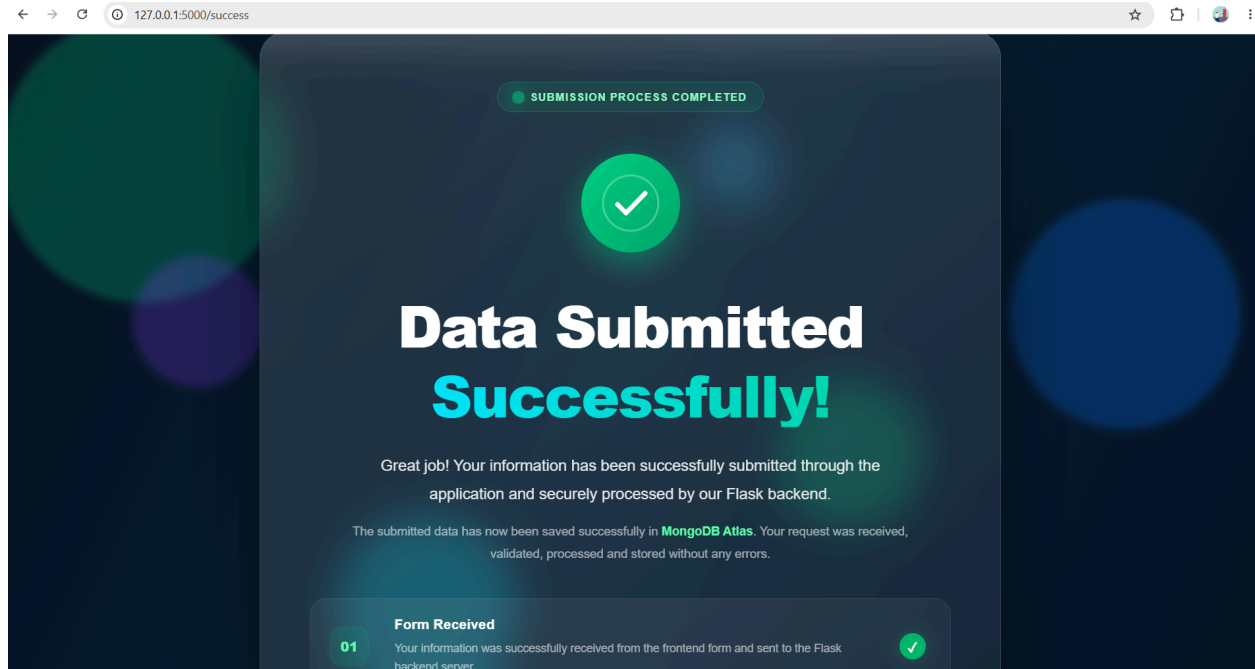
Email Address *

Professional Role *

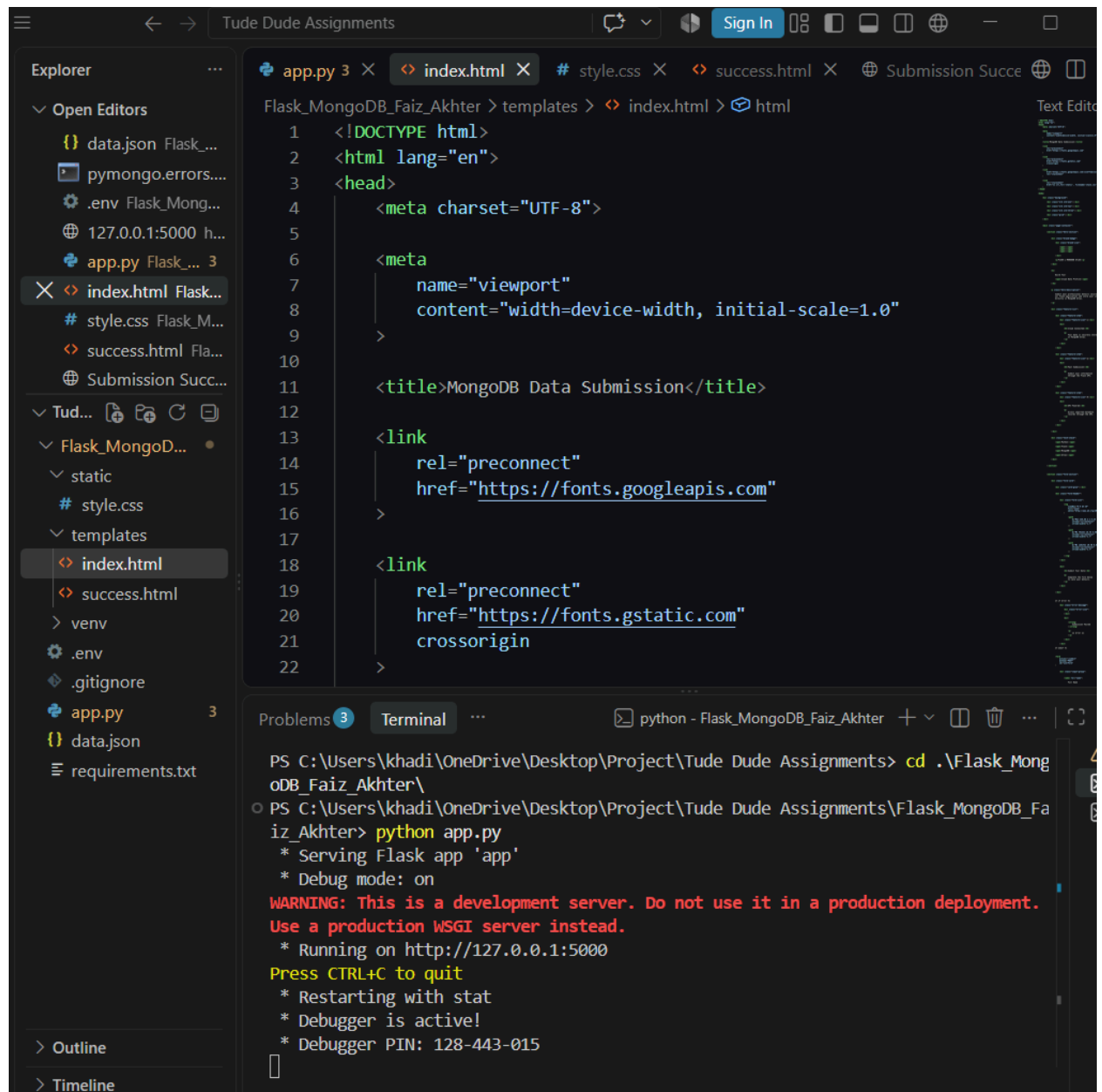
Submit Data →

🔒 Your data is securely processed and stored in MongoDB Atlas.

This is home page



This is success page



index.html page code

Now going to submission of date

FLASK x MONGODB ATLAS

Build Your Cloud Data Profile

Submit your professional details securely through our Flask application and store your information directly in MongoDB Atlas.

✓

Cloud Connected
Your data is securely stored in MongoDB Atlas.

⚡

Fast Submission
Submit your information through the Flask API.

🔗

API Powered
Access imported database records through the API.

🗄️

Submit Your Data
Complete the form below to save your details.

Full Name *

👤 Faiz

Email Address *

✉️ faiz@gmail.com

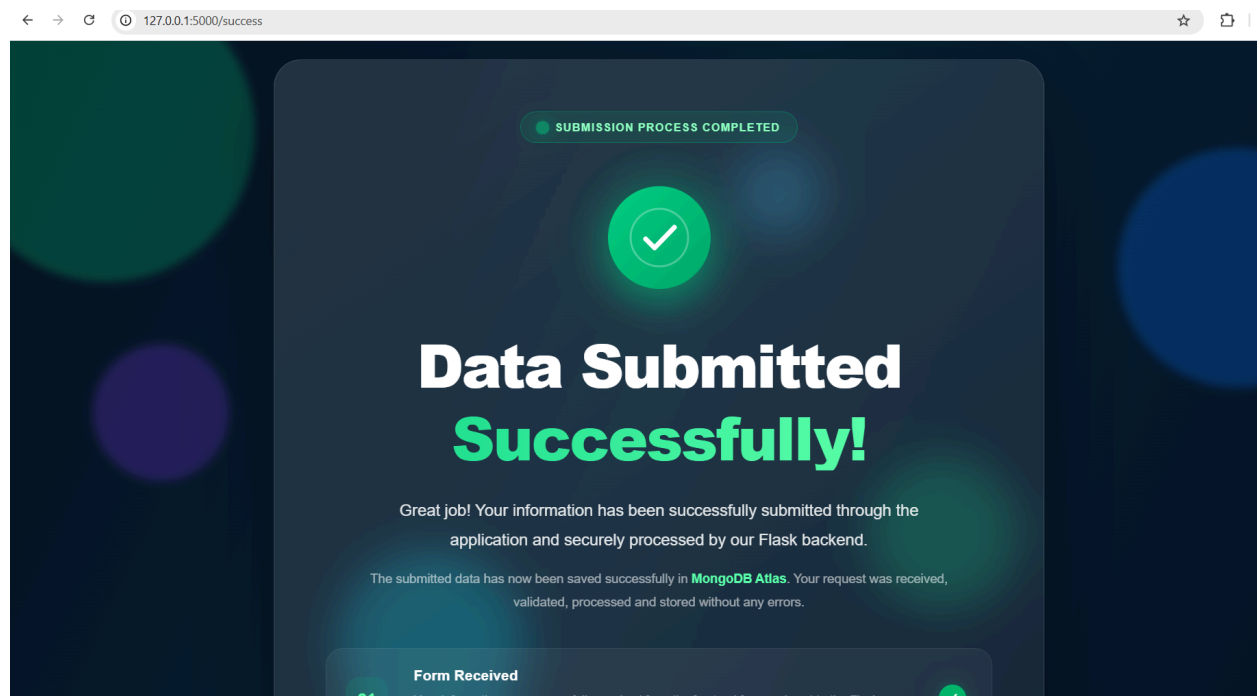
Professional Role *

👤 DevOps Engineer

Submit Data →

🔒

Your data is securely processed and stored in MongoDB Atlas.



data is success fully submitted!!!!

03

Your data has been successfully inserted into the MongoDB Atlas cloud database and is now available through the API.



CLOUD DATABASE STATUS

MongoDB Atlas Connected

Your application is successfully communicating with MongoDB Atlas and your data has been stored securely in the cloud database.

● **CONNECTED**

✓

PROCESS COMPLETED SUCCESSFULLY

Submit Another Entry →

View Submitted Data ↗

Your submission journey has been completed successfully.

Now click on view submitted data

← → ↻ ⓘ 127.0.0.1:5000/api

pretty-print ☐

```
[
  "count": 6,
  "data": [
    {
      "id": 1,
      "name": "Faiz Akhter",
      "role": "DevOps Engineer"
    },
    {
      "id": 2,
      "name": "John Doe",
      "role": "Software Engineer"
    },
    {
      "id": 3,
      "name": "Jane Smith",
      "role": "Cloud Engineer"
    },
    {
      "email": "faizakhter47@gmail.com",
      "name": "faiz khan",
      "role": "devops"
    },
    {
      "email": "Khadija777@gmail.com",
      "name": "Khadija Khatoon",
      "role": "Teacher"
    },
    {
      "email": "faiz@gmail.com",
      "name": "Faiz",
      "role": "DevOps Engineer"
    }
  ],
  "message": "Data fetched successfully"
}
```

So, here you can see in the last line the data is submitted successfully!!!!



The screenshot shows the MongoDB Compass interface. On the left, a sidebar displays the database structure: 'flask_mongodb_db' containing a collection 'users'. The main panel shows the 'users' collection with three documents. The first document has a red ObjectId, name 'faiz khan', email 'faizakhter47@gmail.com', and role 'devops'. The second document, highlighted with a grey background, has a red ObjectId, name 'Khadija Khatoon', email 'Khadija777@gmail.com', and role 'Teacher'. The third document has a red ObjectId, name 'Faiz', email 'faiz@gmail.com', and role 'DevOps Engineer'. Each document has icons for edit, share, and delete on its right side.

```
_id: ObjectId('6a89afd0887aac64fd8dda5c')
name: "faiz khan"
email: "faizakhter47@gmail.com"
role: "devops"

_id: ObjectId('6a89b1a6887aac64fd8dda5d')
name: "Khadija Khatoon"
email: "Khadija777@gmail.com"
role: "Teacher"

_id: ObjectId('6a89b4a2887aac64fd8dda5e')
name: "Faiz"
email: "faiz@gmail.com"
role: "DevOps Engineer"
```

It is successfully visible in Mongo db